Graphical Evaluation of Discrete-Time Convolution

Copyright © Barry Van Veen 2014

AllSignalProcessing.com

Key Concepts

1) Understanding how to perform convolution graphically is useful for developing insight into system behavior.

2) Computers are almost always faster at numerically evaluating discrete-time convolution.

3) The graphical approach proceeds in three steps

$$y[n] = \sum_{k=-\infty}^{\infty} x[n-k]h[k]$$

a) Form $x[n-k]$ by treating n as a constant and k as the independent variable.

   i) Graph $x[k]$

   ii) Reflect $x[k]$ about $k = 0$ to obtain $x[-k]$.

   iii) Shift $x[-k]$ by n to obtain $x[n-k]$. This puts the value that was originally at $k = 0$ at $k = n$. For $n < 0$ shift $x[-k]$ to the left and for $n > 0$ shift $x[-k]$ to the right.

b) Form $x[n-k]h[k]$ as a function of the independent variable k for a given value of n.

c) Sum all the values in $x[n-k]h[k]$ (over k) to obtain $y[n]$.

4) Repeat the three step procedure for each value of n that is of interest.

5) Common types of intuition that can be obtained with the graphical approach:

a) When is the output of the system zero?

b) Does the output increase, decrease, or oscillate?

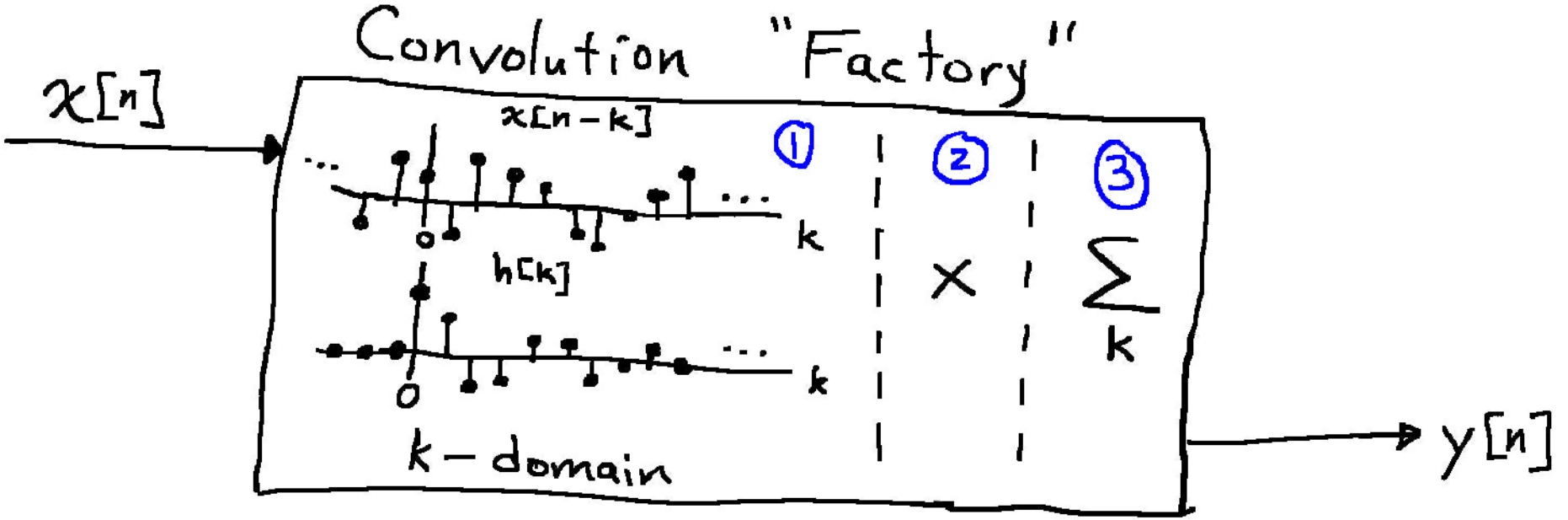
c) Does the output approach a steady-state?

Why ?

- Insight into system behavior
- Computers always win for computation

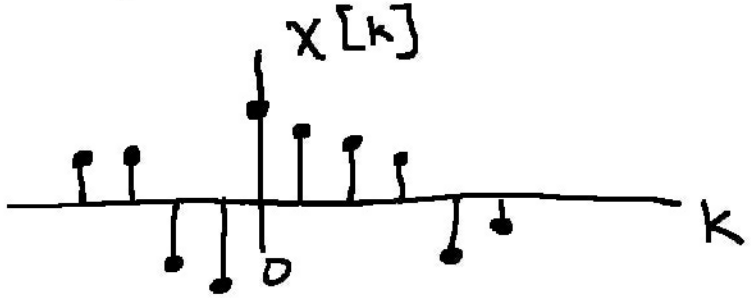
$$y[n] = \sum_{k=-\infty}^{\infty} x[n-k]\,h[k]$$

1 2 3



Step 1: Form $x[n-k]$ as a function of k (n is "constant")

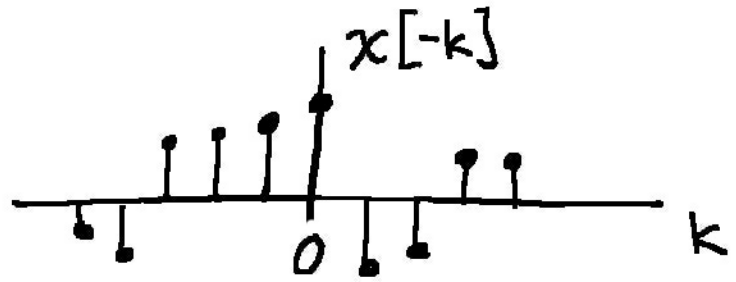
a)



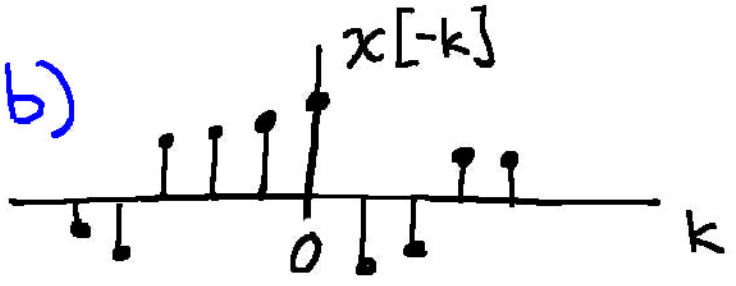
replace k by $-k$

$x[k] \rightarrow x[-k]$

reflect $x[k]$ about $k=0$



b)

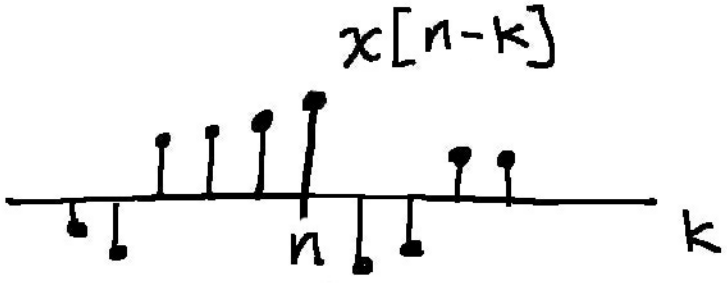


replace k by $k-n$

$x[-k] \rightarrow x[-(k-n)]$

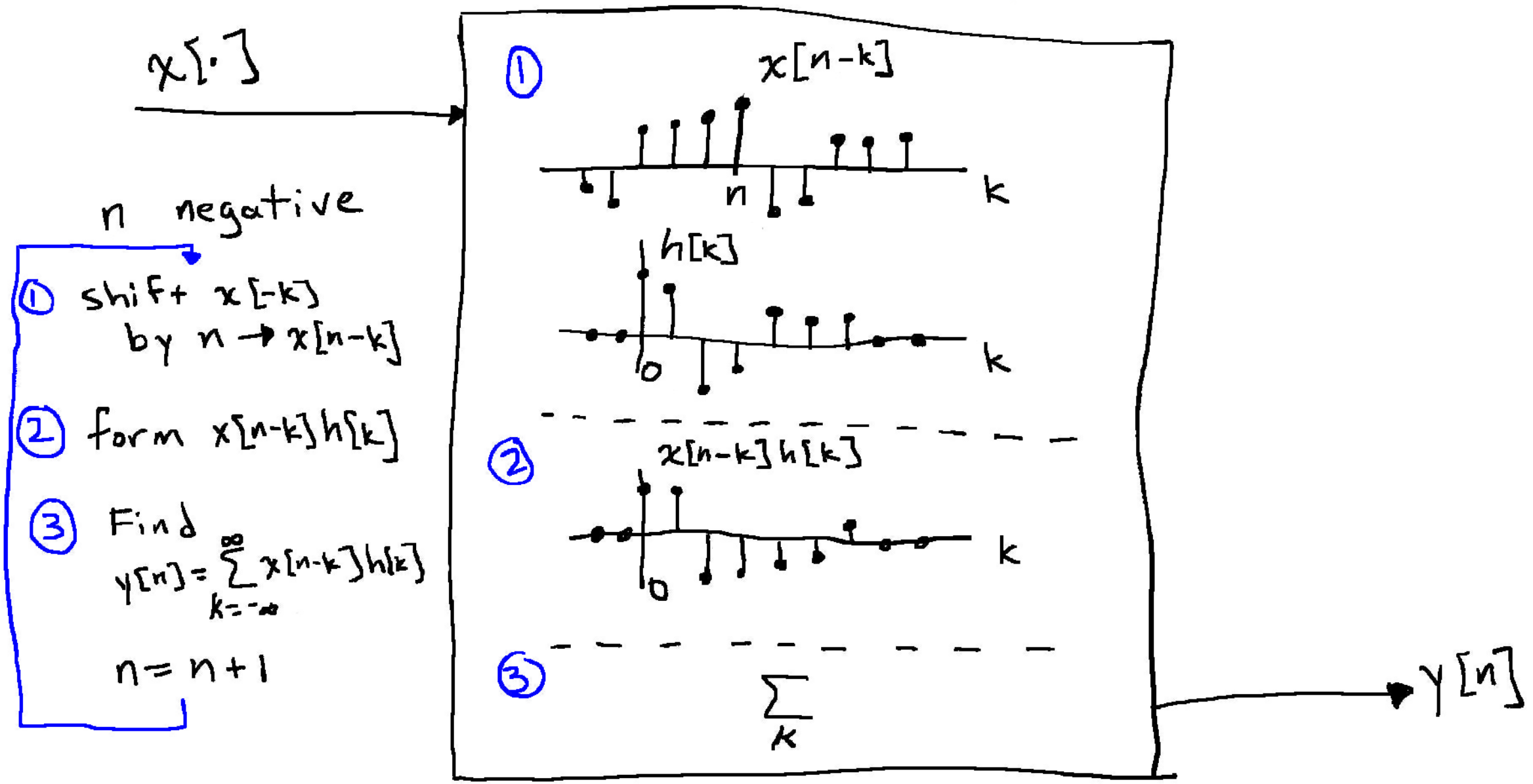
$= x[n-k]$

shift $x[-k]$ right by n steps



Conveyor belt analogy

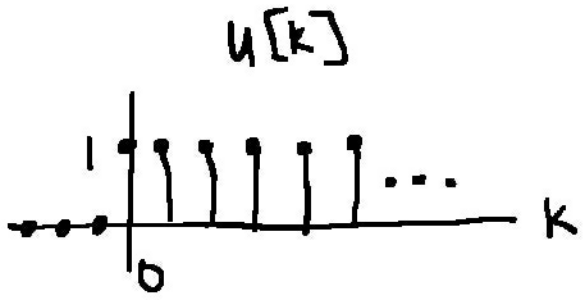
Convolution Factory

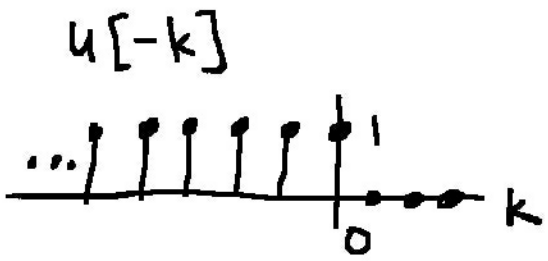


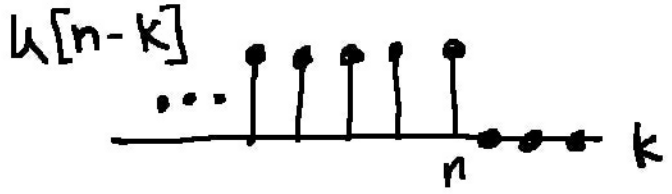
Example: Find $y[n] = h[n] * x[n]$

$x[n] = u[n]$

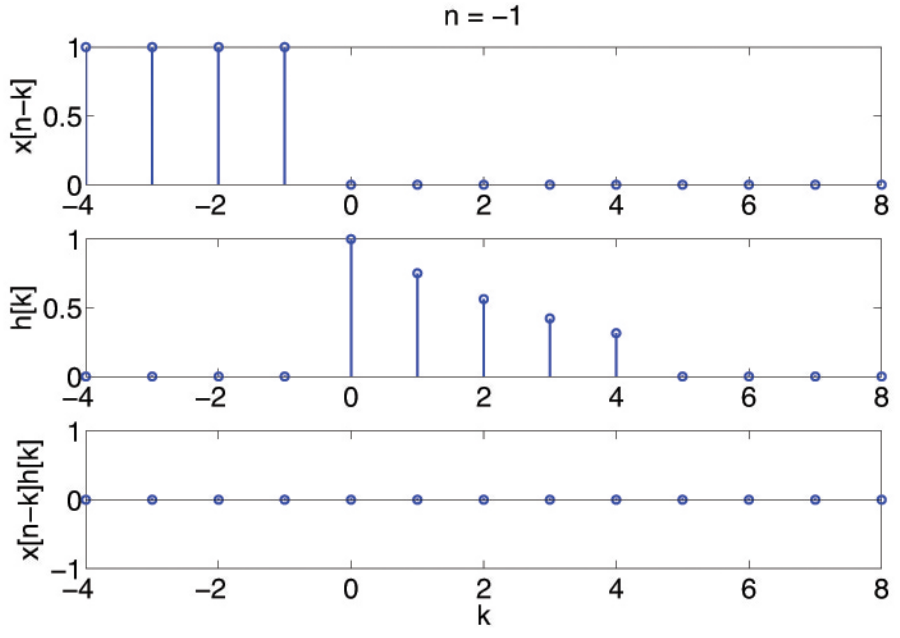
$h[n] = (3/4)^n \{u[n] - u[n-5]\}$







output is 0 for $n < 0$

① 

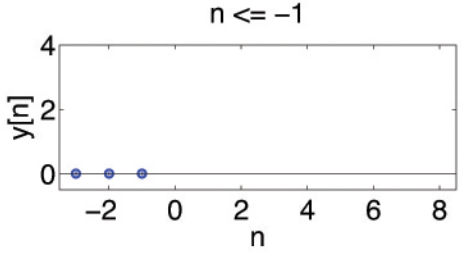
$n < 0$

$x[n-k]\,h[k] = 0$

$\Rightarrow y[n] = 0$

②

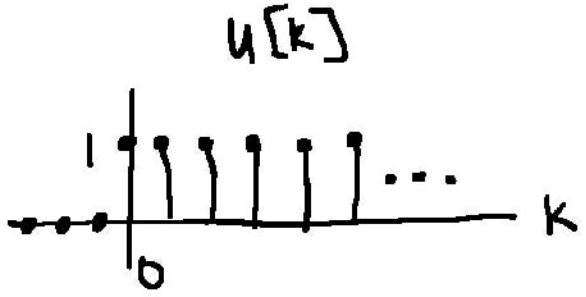
③ $y[-1] = \sum_{k=-\infty}^{\infty} x[-1-k]\,h[k] = 0$

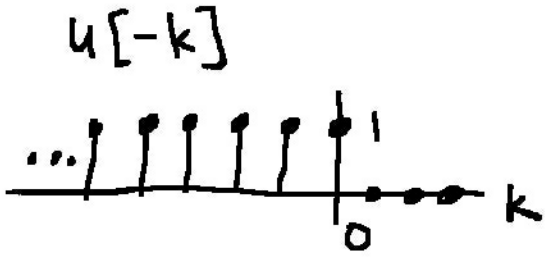


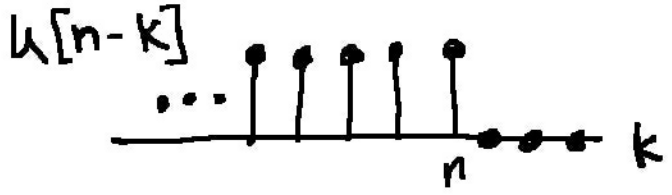
Example: Find $y[n] = h[n] * x[n]$

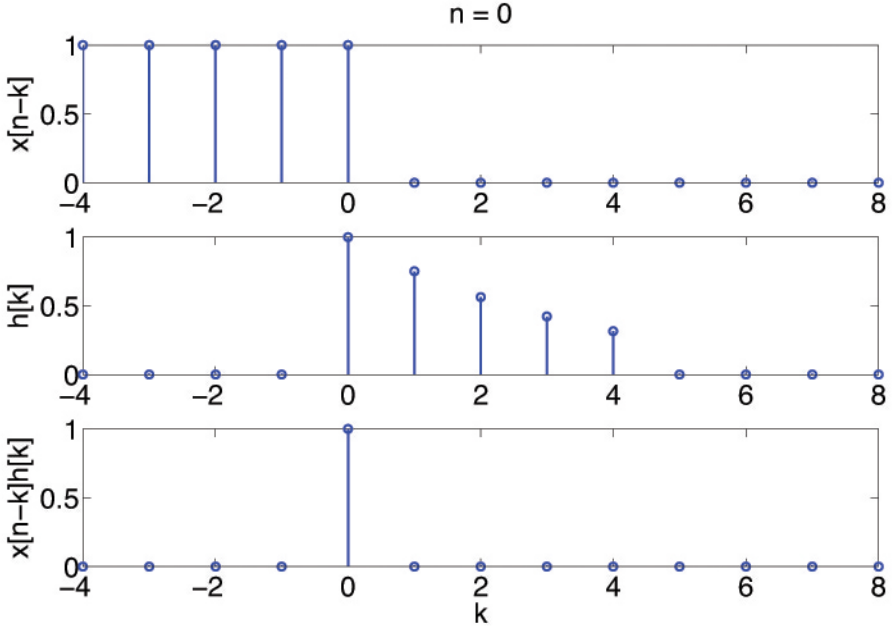
$x[n] = u[n]$

$h[n] = (3/4)^n \{u[n] - u[n-5]\}$

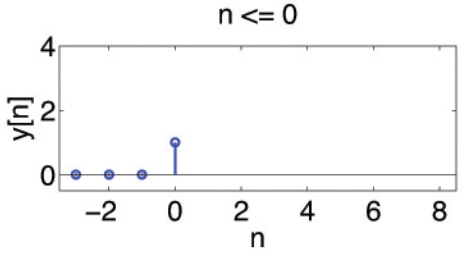






① 

$$x[0-k]h[k] = \begin{cases} 1 & k=0 \\ 0 & k \neq 0 \end{cases}$$

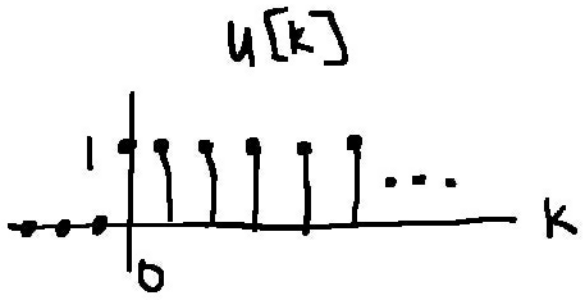
② 

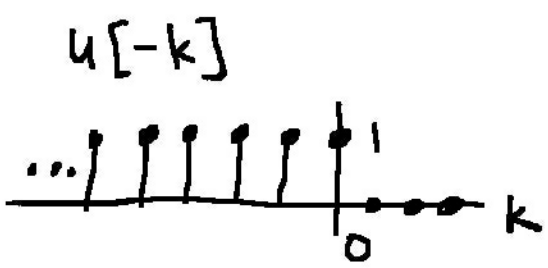
③ $$y[0] = \sum_{k=-\infty}^{\infty} x[0-k]h[k] = 1$$

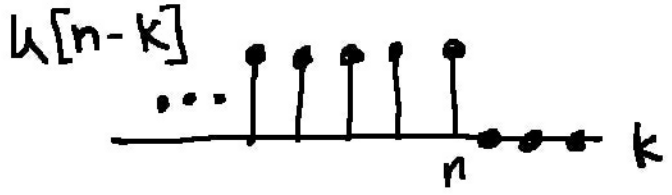
Example: Find $y[n] = h[n] * x[n]$

$x[n] = u[n]$

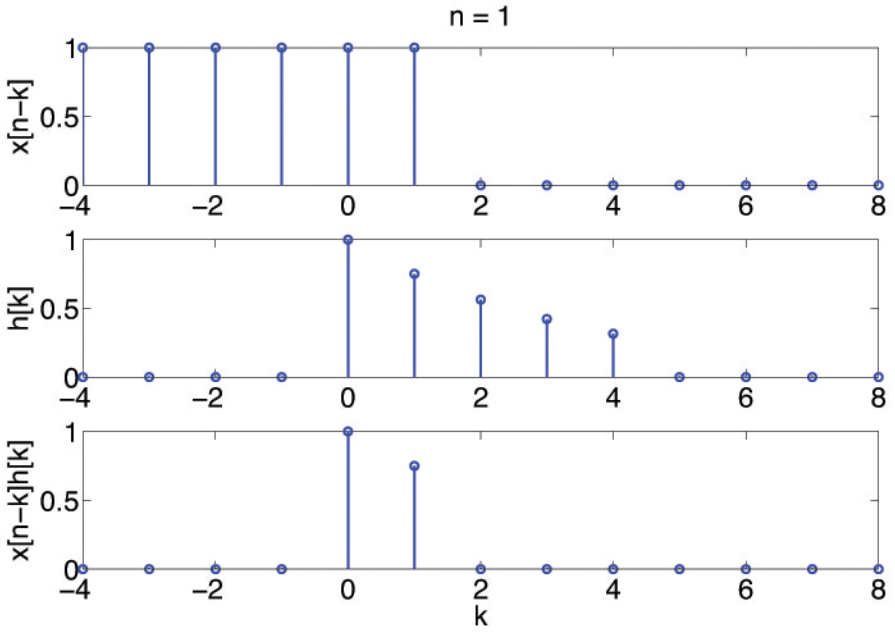
$h[n] = (3/4)^n \{u[n] - u[n-5]\}$







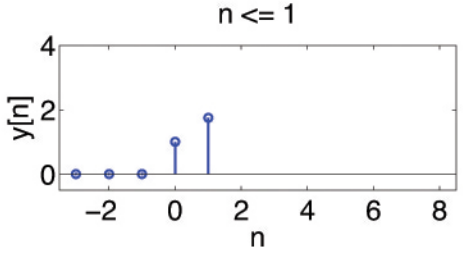
output increasing for $n > 0$ ①



② $n = 1$

$$x[1-k]h[k] = \begin{cases} (3/4)^k & k = 0, 1 \\ 0 & k < 0, k > 1 \end{cases}$$

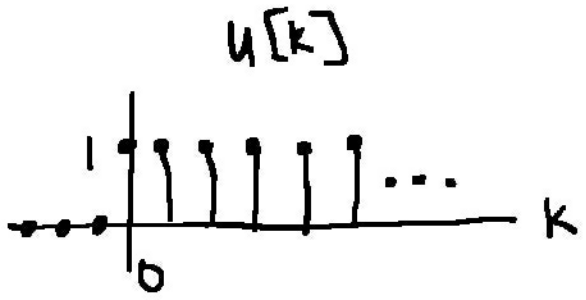
③ $$y[1] = \sum_{k=-\infty}^{\infty} x[1-k]h[k] = 1.75$$

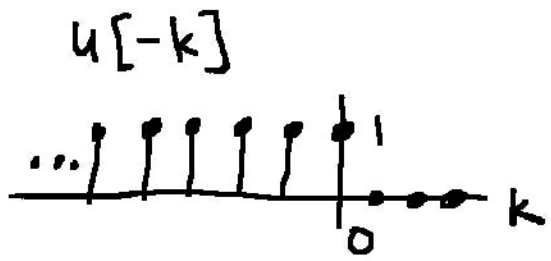


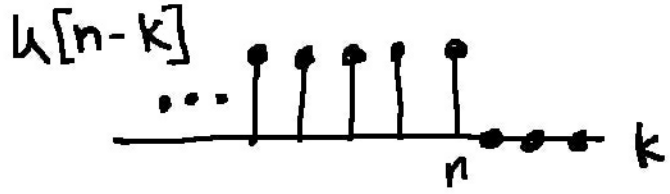
Example: Find $y[n] = h[n] * x[n]$

$x[n] = u[n]$

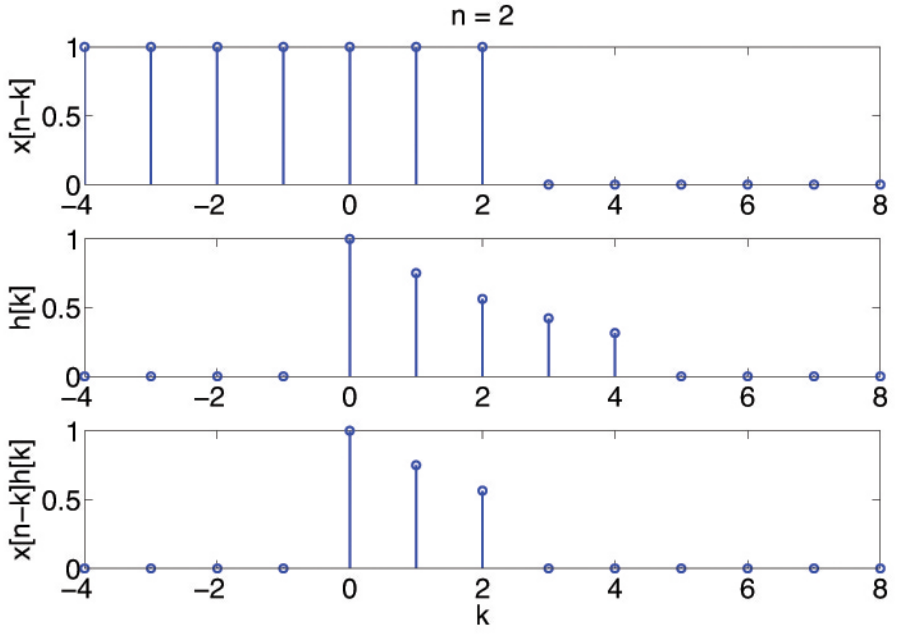
$h[n] = (3/4)^n \{u[n] - u[n-5]\}$







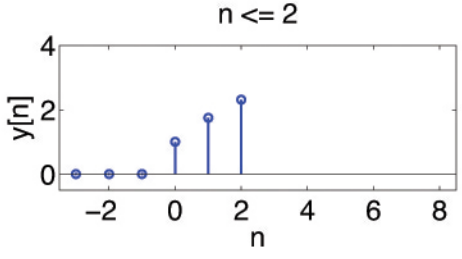
output continues to increase until $n = 4$ ①



② $0 \le n \le 4$

$$x[n-k]\,h[k] = \begin{cases} (3/4)^k & 0 \le k \le n \\ 0 & k < 0,\ k > n \end{cases}$$

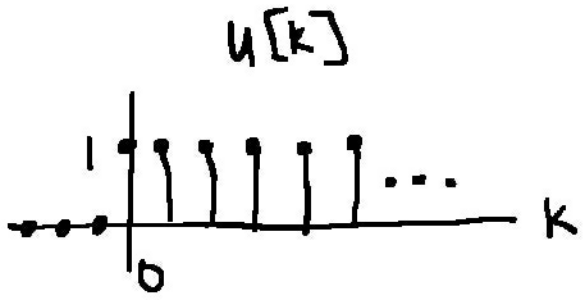
③ $$y[2] = \sum_{k=-\infty}^{\infty} x[2-k]\,h[k] = 2.31$$

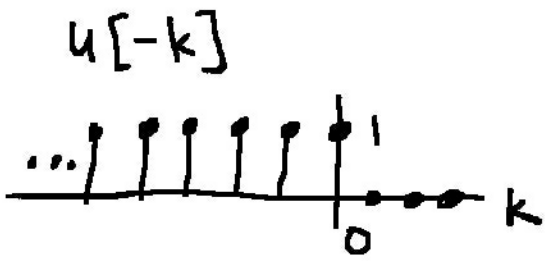


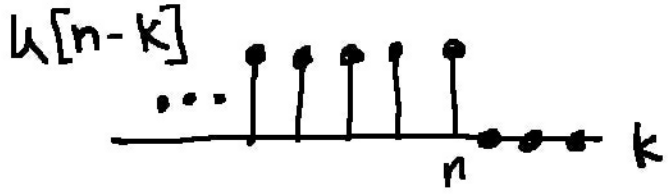
Example: Find $y[n] = h[n] * x[n]$

$x[n] = u[n]$

$h[n] = (3/4)^n \{u[n] - u[n-5]\}$

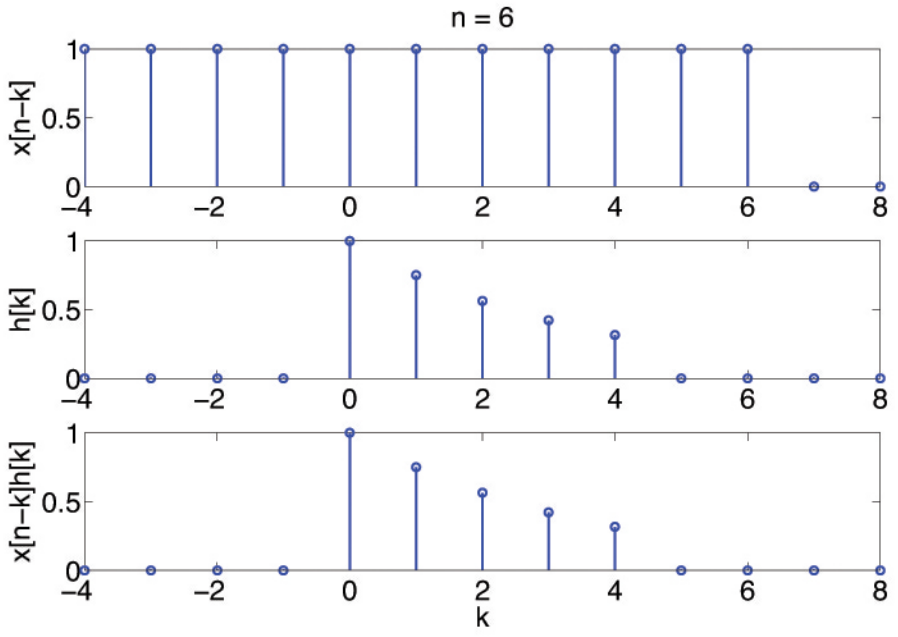






output constant for $n \geq 4$

① ② ③

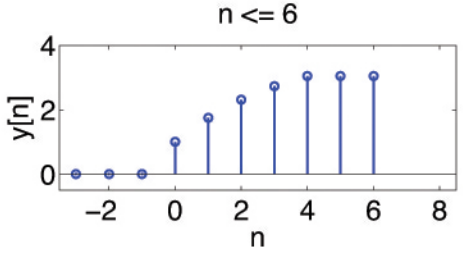


$n > 4$

$$x[n-k]h[k] = \begin{cases} (3/4)^k & 0 \leq k \leq 4 \\ 0 & k<0, k>4 \end{cases}$$

$$y[n] = \sum_{k=0}^{4} (3/4)^k$$

$$y[6] = \sum_{k=-\infty}^{\infty} x[6-k]h[k] = 3.05$$



Copyright 2014
Barry Van Veen